| Problem Chosen | 2026 Mathematical Modeling Summary Sheet | Team Control Number |
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# From Private Incentives to Public Stability: Modeling Free Riding in Dynamic Public-Good Provision


## Summary

Public goods such as environmental governance, open-source maintenance, public infrastructure, and community services create a common governance dilemma: the collective stock benefits all participants, while contribution costs are borne privately. This paper develops a simulation-based *Dynamic Stock–Pressure Free-Riding* (DSPF) framework to explain how decentralized individual rationality can generate persistent under-provision and how policy instruments can mitigate the resulting welfare loss.

The model combines heterogeneous agents, a dynamic public-good stock, maintenance pressure, demand feedback, capacity saturation, and policy intervention. Agents differ in benefit coefficients, contribution costs, contribution efficiencies, pressure sensitivities, and initial behavioral types. Four controlled synthetic scenarios are constructed: small volunteer provision, rapid growth, critical infrastructure, and burnout-prone provision. All numerical inputs and outputs are explicitly labeled as synthetic or simulation-based, and the random seed is fixed at 20260614 for reproducibility.

We compare two decision benchmarks under identical state conditions. The first is a Nash-style individual-rational solution computed by damped marginal-response updates. The second is a stage-wise social-planner benchmark solved by numerical optimization. Under the updated capacity-saturating dynamics, simulation results indicate a systematic free-riding gap of approximately 0.805–0.860, an under-provision ratio of approximately 0.226–0.299, and a welfare-loss ratio of approximately 0.101–0.198. The burnout-prone scenario is the most severe welfare-loss case because high cost and maintenance pressure jointly amplify the loss from insufficient contribution.

We then test subsidy, penalty, reputation, matching fund, threshold governance, and combined portfolio policies. In the critical-infrastructure scenario, the baseline long-run welfare is 66.52. Subsidy, penalty, reputation, threshold governance, and combined portfolio policies improve welfare and reduce free riding. The combined portfolio produces the highest long-run welfare and the largest pressure reduction, while reputation produces nearly the same welfare at much lower policy cost. Pareto, response-surface, robustness, and Monte Carlo experiments further show that welfare, policy cost, pressure relief, fairness, and free-riding reduction cannot generally be optimized by one instrument alone.

The model suggests three governance principles: use low-cost incentive correction before heavy intervention, switch to portfolio governance when maintenance pressure is high, and select policies by scenario-specific trade-offs rather than by a single welfare score. These conclusions are mechanism-oriented and conditional on the synthetic simulation design, not empirical estimates for a specific public-good system.



**Keywords**: Public Goods; Free Riding; Dynamic Game; Social-Planner Benchmark; Policy Intervention; Simulation-Based Modeling

# Contents

# 1 Introduction

## 1.1 Background

Public goods have two important characteristics: the benefit of provision is shared by many agents, while the cost of contribution is usually borne by the contributor. Classical public-good theory explains why private provision may be inefficient when benefits are non-excludable and shared across agents [1], [2], [3]. Environmental governance, open-source software maintenance, public infrastructure, and community services all exhibit this structure. When agents can enjoy the outcome even if they contribute little, each agent has an incentive to wait for others to contribute. This strategic incentive may create a free-riding gap between decentralized individual behavior and socially desirable provision.

The difficulty is inherently dynamic. Public-good quality is not produced once and then fixed. It decays naturally, is affected by demand and maintenance pressure, and may improve or deteriorate over time depending on repeated contribution. A static model can show why private incentives are insufficient, but it cannot show how under-provision accumulates, how maintenance pressure feeds back into future quality, or why different policies have different long-run effects. The commons-governance literature also suggests that institutions, monitoring, sanctions, and local rules may change contribution behavior rather than merely changing the physical stock [4], [5]. Therefore, we model the problem as a dynamic multi-agent system with heterogeneous agents and repeated contribution decisions.

## 1.2 Restatement of the Problem

The problem asks us to establish a multi-agent dynamic supply model for public goods. The model should describe the stock of a public good over time, include individual contribution, natural decay, individual benefit, contribution cost, and free-riding behavior, compare individual-rational provision with a stage-wise social-planner benchmark, and then design possible interventions and management recommendations. The emphasis is on mechanism-oriented mathematical modeling. Real data fitting is not required; reasonable synthetic parameter settings, parameter experiments, and sensitivity analysis are acceptable as long as their assumptions are stated.

Accordingly, we answer four questions:

1. How can heterogeneous agents and dynamic public-good stock be represented in one model?
2. Does Nash-style individual rationality produce lower contribution than a social-planner benchmark under controlled scenario settings?
3. How large are the free-riding gap, under-provision ratio, welfare loss, and maintenance pressure under different synthetic scenarios?
4. Which policy instruments can reduce free riding without creating excessive policy cost, inequality, or long-run instability?

## 1.3 Our Work

We organize the paper around a simulation-based *Dynamic Stock–Pressure Free-Riding* (DSPF) framework. The framework treats contribution incentives, public-good stock, maintenance pressure, and policy instruments as one coupled system. We do not claim that the synthetic scenarios are empirical observations. Instead, the scenarios are controlled numerical testbeds designed to examine the mechanism of dynamic free riding. Figure 1 summarizes the DSPF modeling framework.

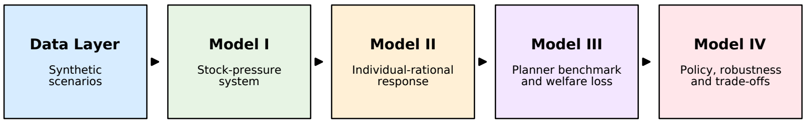


**Figure 1:** Simulation-based modeling framework. Synthetic scenarios generate heterogeneous agents and dynamic states. Nash-style individual rationality is compared with a social-planner benchmark, and policy interventions are evaluated by welfare, cost, free-riding reduction, fairness, maintenance pressure, and stability.

Figure 1 is used as the organizing logic of the paper. The left part of the diagram defines the controlled input space: synthetic scenarios and heterogeneous agents. The middle part describes the model mechanism: individual efforts are converted into effective contribution, which changes public-good stock and maintenance pressure over time. The right part contains the decision and evaluation layer: decentralized individual rationality is compared with the social-planner benchmark, and policy interventions are evaluated by several long-run indicators. This organization clarifies the modeling route before the detailed equations are introduced, so the later numerical experiments can be traced back to specific mechanisms rather than to isolated simulations. Model I answers how the public-good stock and pressure evolve, Model II explains how decentralized agents choose effort under private incentives, Model III measures the welfare loss created by this decentralization, and Model IV compares policy instruments that can correct the marginal-incentive gap.

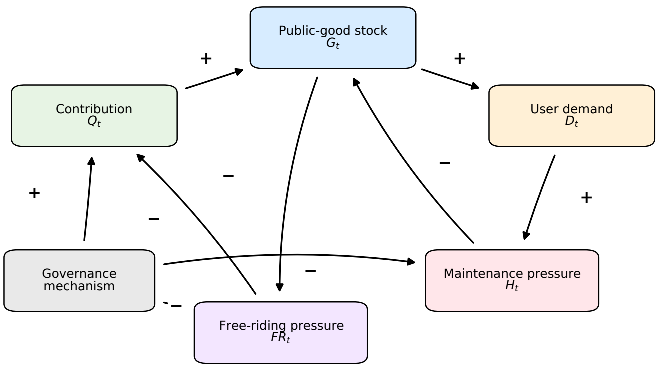


**Figure 2:** Causal feedback structure of the DSPF framework. The diagram separates the physical stock-pressure dynamics from the behavioral free-riding channel and the policy-intervention channel; edge signs indicate positive or negative causal effects.

A second mechanism view is shown in Figure 2. Unlike the workflow diagram, the causal-loop diagram emphasizes feedback: contribution improves stock and relieves pressure, higher stock attracts demand, demand may rebuild maintenance pressure, and pressure can discourage sustainable provision if private incentives remain weak. This feedback interpretation motivates why our numerical experiments report not only contribution and welfare, but also pressure, fairness, policy cost, and stability.

The main contributions are as follows:

- We construct a DSPF stock-pressure-demand model for public-good provision with heterogeneous agents.
- We compute a Nash-style decentralized solution and a stage-wise social-planner benchmark under identical synthetic scenarios.
- We quantify free-riding gap, under-provision ratio, welfare loss, maintenance pressure, fairness, and stability.
- We compare six intervention policies and a baseline under multi-criteria evaluation.
- We conduct sensitivity, robustness, Monte Carlo, and Pareto trade-off analyses to avoid over-interpreting one parameter setting.

# 2 Assumptions and Notations

## 2.1 Assumptions and Justifications

**Assumption 1: Synthetic scenarios are mechanism-oriented testbeds.** The four scenarios are stylized numerical environments, not real-world observations. This is consistent with the problem requirement that the emphasis is on mechanisms, parameter experiments, and sensitivity analysis rather than empirical prediction.

**Assumption 2: Agents are heterogeneous but boundedly simple.** Each agent has a benefit coefficient, contribution cost, contribution efficiency, pressure sensitivity, and initial behavioral type. These variables capture the main differences needed for a public-good model, while keeping the system interpretable.

**Assumption 3: Public-good quality is represented by a normalized stock.** The public-good stock is not measured in physical units. It represents the relative quality or service level of a public good, such as infrastructure reliability, software maintainability, or environmental quality.

**Assumption 4: The social-planner benchmark is stage-wise.** The social planner maximizes current-period welfare given the current state and the next-state transition. We do not claim to solve an infinite-horizon dynamic programming problem.

**Assumption 5: Nash-style individual rationality is a computational equilibrium proxy.** We use damped iterative marginal-response updates to obtain an approximate fixed point of individual-rational decisions. We do not prove uniqueness of an analytical Nash equilibrium.

**Assumption 6: Policy cost is explicitly counted.** Interventions are not free. Subsidy, matching fund, reputation operation, public budget injection, backlog reduction, and penalty administration all enter the policy cost function.

## 2.2 Notations

To keep the notation readable, we separate state and agent variables from policy and evaluation variables. The separation also reflects the logic of the numerical pipeline: first the agent and state variables determine the decentralized effort response, then the policy variables perturb marginal incentives or state transitions, and finally the evaluation variables summarize the simulated outcome. This ordering prevents the model from treating policy performance indicators as primitive assumptions.

**Table 1:** State and agent notations used in the dynamic public-good model.

| Symbol | Meaning |
|---|---|
| $N, t$ | Number of agents and discrete time index |
| $e_{i,t}, e_{\max}$ | Effort of agent $i$ and the maximum feasible effort |
| $a_i, b_i, c_i, \mu_i$ | Contribution efficiency, benefit coefficient, cost coefficient, and pressure sensitivity of agent $i$ |
| $Q_t, Q_{\max}, q(Q_t)$ | Aggregate contribution, maximum aggregate contribution, and normalized effective contribution |
| $G_t, H_t, D_t$ | Public-good stock, maintenance pressure, and demand level at time $t$ |
| $\bar{G}, \bar{H}, \bar{D}$ | Scenario-specific stock, pressure, and demand capacity bounds used in saturation and projection |
| $\delta, \alpha, \eta, \eta_G, \eta_H$ | Natural stock decay, contribution-to-stock conversion, pressure damage, stock-saturation power, and pressure-saturation power |
| $\rho, \lambda, \kappa$ | Natural pressure dissipation, demand-to-pressure conversion, and contribution-based pressure relief |
| $g_D, \chi, \psi, \varepsilon_t$ | Demand growth, stock attraction, pressure deterrence, and simulation noise |
| $V(G), \omega$ | Concave public-good value function and its scaling parameter |

**Table 2:** Policy and evaluation notations.

| Symbol | Meaning |
|---|---|
| $s, r, p, m, B, R, \tau$ | Subsidy, reputation, penalty, matching fund, budget injection, backlog reduction, and threshold parameters |
| $\pi, C_{\text{policy}}$ | Policy vector and policy cost |
| $F_i, FR, \bar{b}_q, \epsilon$ | Free-rider indicator, free-riding ratio, high-benefit cutoff, and low-effort threshold |
| $W_t, \mathbf{e}_t^{SO}$ | Social welfare and stage-wise social-planner effort vector |
| $\Delta FR, \Delta H$ | Reduction in free-riding ratio and maintenance pressure relative to the baseline |
| $\text{Gini}(e)$ | Inequality measure of the individual effort distribution |

The notation system is arranged to keep the model hierarchy explicit. State variables describe the evolving public-good environment, agent variables describe heterogeneous private incentives, policy variables describe external intervention, and evaluation variables describe the criteria used for comparison. This separation helps ensure that later tables do not mix model inputs, decision variables, and performance outputs. In particular, $Q_t$, $G_t$, and $H_t$ are dynamic outcomes generated by effort decisions, while $s$, $p$, $r$, $m$, $B$, and $R$ are policy levers used only in the intervention experiments.

# 3 Data Strategy and Simulation-Based Scenario Design

## 3.1 Synthetic-Only Data Strategy

This paper uses a synthetic-only data route because the course problem emphasizes mechanism-oriented modeling rather than empirical forecasting. Agent attributes, scenario parameters, and trajectories are generated as controlled numerical testbeds. The goal is therefore not to estimate a particular community, city, platform, or infrastructure system, but to reveal how decentralized free riding can create dynamic under-provision under transparent assumptions.

The numerical design is fully specified by the fixed random seed 20260614, four scenarios, seven policies, a 30-period simulation horizon, 50 Monte Carlo runs, and 80 random policy samples for Pareto analysis. Thus the numerical results can be reproduced from the stated scenario and policy settings. In the interpretation of the results, welfare, pressure, and free-riding ratios should be read as controlled simulation outputs rather than observed real-world magnitudes. This boundary is important: the paper uses simulation to study mechanisms, not to estimate a specific community, city, platform, or infrastructure system.

## 3.2 Heterogeneous Agent Generation

For each scenario, $N = 35$ agents are generated. Agent benefits are sampled from a log-normal distribution, contribution costs from a gamma distribution, contribution efficiencies from a uniform distribution, and pressure sensitivities from a uniform distribution. Scenario-specific benefit and cost scales adjust the difficulty of provision.

$$
\begin{aligned}
b_i &\sim \mathrm{LogNormal}(\mu_b, \sigma_b) \cdot s_b, \\
c_i &\sim \mathrm{Gamma}(k_c, \theta_c) \cdot s_c, \\
a_i &\sim U(a_{\min}, a_{\max}), \\
\mu_i &\sim U(\mu_{\min}, \mu_{\max}).
\end{aligned}
$$

Initial behavioral types are sampled from three categories: active contributor, conditional contributor, and passive user. These types are not fixed strategies in the equilibrium computation, but they provide an interpretable description of the heterogeneous population.

## 3.3 Synthetic Scenario Design

Table 3 lists the major parameters of the four synthetic scenarios. The small-volunteer scenario describes a low-pressure voluntary contribution system. The rapid-growth scenario represents increasing demand. The critical-infrastructure scenario has higher public-good importance and maintenance pressure. The burnout-prone scenario combines high cost and high pressure, making under-provision more likely.

**Table 3:** Synthetic scenario parameters.

| Scenario | $G_0$ | $H_0$ | $D_0$ | $\delta$ | $\alpha$ | $\eta$ | $\lambda$ | $\kappa$ | Benefit scale | Cost scale |
|---|---|---|---|---|---|---|---|---|---|---|
| Small volunteer | 0.58 | 0.20 | 0.30 | 0.020 | 0.72 | 0.16 | 0.18 | 0.52 | 0.80 | 0.90 |
| Rapid growth | 0.60 | 0.28 | 0.42 | 0.038 | 0.66 | 0.24 | 0.40 | 0.34 | 1.05 | 1.03 |
| Critical infrastructure | 0.62 | 0.34 | 0.58 | 0.050 | 0.78 | 0.30 | 0.44 | 0.38 | 1.35 | 1.08 |
| Burnout-prone | 0.56 | 0.46 | 0.48 | 0.060 | 0.58 | 0.36 | 0.42 | 0.24 | 1.00 | 1.45 |

*These are controlled numerical settings, not empirical observations. Scenario-specific capacities are used in the simulation; the stock- and pressure-saturation powers are fixed at $\eta_G = 2.0$ and $\eta_H = 1.0$.*

The parameter design is intended to separate mechanisms rather than to fit a particular public-good system. Small-volunteer provision uses low initial pressure and moderate cost to represent a relatively benign voluntary environment. Rapid growth increases demand and demand-to-pressure conversion, so delayed pressure accumulation becomes visible. Critical infrastructure raises the benefit scale and pressure damage because service failure is more costly. Burnout-prone provision combines a high cost scale, high initial pressure, and weaker pressure relief, representing systems in which repeated contribution is difficult to sustain.

# 4 Model I: Dynamic Stock–Pressure System

## 4.1 Aggregate Effective Contribution

Each agent chooses a contribution effort $e_{i,t} \in [0, e_{\max}]$. The aggregate effective contribution is

$$Q_t = \sum_{i=1}^{N} a_i e_{i,t}.$$

Because public-good production often has diminishing returns, we use a logarithmic transformation

$$q(Q_t) = \frac{\log(1+Q_t)}{\log(1+Q_{\max})},$$

where $Q_{\max} = \sum_i a_i e_{\max}$. Thus $q(Q_t)$ is a normalized contribution effectiveness term.

## 4.2 Public-Good Stock Dynamics with Capacity Saturation

The public-good stock evolves through a capacity-constrained saturating transition. First define the stock saturation factor

$$\Gamma_G(G_t) = 1 - \left(\frac{G_t}{\bar{G}}\right)^{\eta_G}, \qquad 0 \le G_t \le \bar{G}.$$

The unprojected next-period stock is

$$\widetilde{G}_{t+1} = (1-\delta)G_t + \alpha(1+m)q(Q_t)\Gamma_G(G_t) - \eta H_t + B, \tag{1}$$

and the feasible stock is

$$G_{t+1} = \Pi_{[0,\bar{G}]}(\widetilde{G}_{t+1}). \tag{2}$$

Here $\delta$ is natural decay, $\alpha$ is the conversion rate from contribution to stock, $m$ is matching-fund intensity, $\eta$ measures damage from maintenance pressure, $B$ is direct public budget injection, and $\Pi$ denotes projection to the feasible interval. The saturation factor is the main economic mechanism: when the public-good stock approaches its scenario-specific capacity $\bar{G}$, the marginal stock improvement from additional contribution becomes smaller. The final projection is only a feasibility safeguard.

## 4.3 Maintenance Pressure and Demand Dynamics

Maintenance pressure is also modeled as a bounded dynamic state. Let

$$\Gamma_H(H_t) = 1 - \left(\frac{H_t}{\bar{H}}\right)^{\eta_H}, \qquad 0 \le H_t \le \bar{H}.$$

The pressure transition is

$$\widetilde{H}_{t+1} = (1-\rho)H_t + \lambda D_t \Gamma_H(H_t) - \kappa q(Q_t) - R, \tag{3}$$

$$H_{t+1} = \Pi_{[0,\bar{H}]}(\widetilde{H}_{t+1}), \tag{4}$$

where $\rho$ is natural pressure dissipation, $\lambda$ is demand-to-pressure conversion, $\kappa$ is contribution-based pressure relief, and $R$ is direct backlog reduction. Free riding therefore affects pressure through insufficient aggregate contribution and weaker pressure relief, rather than through a separate ad hoc free-riding term.

Demand evolves as

$$D_{t+1} = D_t \left(1 + g_D + \chi \frac{G_t}{\bar{G}} - \psi \frac{H_t}{\bar{H}}\right) + \varepsilon_t, \tag{5}$$

with final projection to the scenario-specific demand range. The term $g_D$ is baseline demand growth. Better public-good stock may attract more demand, while higher maintenance pressure may reduce demand or effective usage. The noise term $\varepsilon_t$ is used only in simulation, and its scale is specified by the synthetic scenario.

**Proposition 1** (Feasible state invariance)**.** Suppose the initial state satisfies $G_0 \in [0, \bar{G}]$, $H_0 \in [0, \bar{H}]$, and $D_0 \in [0, \bar{D}]$. If the stock, pressure, and demand updates are followed by their stated interval projections, then every simulated trajectory satisfies

$$G_t \in [0, \bar{G}], \qquad H_t \in [0, \bar{H}], \qquad D_t \in [0, \bar{D}]$$

for all periods $t$, for any feasible effort vector and any policy vector used in the experiments.

*Proof.* For any closed interval $[L, U]$, the projection operator satisfies $\Pi_{[L,U]}(x) \in [L, U]$ for every real number $x$. Equations 2 and 4 apply this operator to the raw stock and pressure updates; the demand update is treated in the same projected manner in the simulation. Therefore the next state remains inside the feasible interval whenever the current state is simulated. The claim follows by induction over $t$. □

The dynamic equations give two useful local monotonicity relations. Since

$$\frac{\partial q(Q_t)}{\partial e_{i,t}} = \frac{a_i}{(1 + Q_t)\log(1 + Q_{\max})} > 0,$$

we have, at interior points away from projection boundaries,

$$\frac{\partial \widetilde{G}_{t+1}}{\partial e_{i,t}} = \alpha(1 + m)\Gamma_G(G_t)\frac{a_i}{(1 + Q_t)\log(1 + Q_{\max})} \geq 0,$$

$$\frac{\partial \widetilde{H}_{t+1}}{\partial e_{i,t}} = -\kappa \frac{a_i}{(1 + Q_t)\log(1 + Q_{\max})} < 0.$$

Thus individual effort has two system-level effects: it increases next-period public-good stock when capacity is not binding, and it reduces next-period maintenance pressure. These two signs are the technical foundation of the marginal externality derivation in the next model.

## 4.4 Value Function and Welfare Components

Public-good stock creates utility through a concave value function

$$V(G) = \log(1 + \omega G),$$

where $\omega = 3.6$ in the numerical experiments. Concavity reflects diminishing marginal benefit from stock improvement.

Equations 1–5 define the dynamic core of the DSPF framework. They also explain why the problem cannot be reduced to a one-period public-good game: current effort changes both the next stock and the next maintenance burden, and these state variables then affect future welfare and policy needs. The numerical behavior generated by this dynamic core is discussed in Section 9.1.

# 5 Model II: Individual Rationality and Nash-Style Solution

## 5.1 Individual Utility

An individual agent trades off public-good benefit, contribution cost, pressure exposure, and policy incentives. The setup follows the standard game-theoretic idea that each agent chooses a feasible action while treating the other agents' actions and the current state as given [6], [7]. Given next-period stock and pressure, the utility of agent $i$ is

$$u_i(e_i, e_{-i}; \pi) = b_i V(G_{t+1}) - \frac{1}{2} c_i e_i^2 - \mu_i H_{t+1} + s e_i + r \log(1 + e_i) \\ + p \left[ \mathbf{1}_{\{b_i \geq \bar{b}_q\}} \left( 0.35 e_i - 1.4 (\tau^\star - e_i)_+^2 \right) - 0.10 (\tau^\star - e_i)_+ \right].$$

Here $s$ is subsidy intensity, $r$ is reputation intensity, $p$ is penalty intensity, $\bar{b}_q$ is the high-benefit cutoff, and $\tau^\star = \max(\tau, \epsilon)$ is the implemented threshold. This expression matches the implemented synthetic utility: high-benefit agents receive the targeted penalty-related incentive around the threshold, while all below-threshold agents receive a small threshold-push term. The structure directly represents the free-riding incentive: high-benefit agents may obtain public-good benefit even when their own effort is low.

The marginal incentive gap can be seen by comparing the individual and social first-order conditions. Ignoring boundary projection and non-smooth threshold terms for exposition, agent $i$ responds to

$$\frac{\partial u_i}{\partial e_i} \approx b_i V'(G_{t+1}) \frac{\partial G_{t+1}}{\partial e_i} - c_i e_i - \mu_i \frac{\partial H_{t+1}}{\partial e_i} + \text{policy marginal terms}, \tag{6}$$

whereas the social planner internalizes the effect of agent $i$'s effort on all agents:

$$\frac{\partial W}{\partial e_i} \approx \sum_{j=1}^{N} b_j V'(G_{t+1}) \frac{\partial G_{t+1}}{\partial e_i} - c_i e_i - \sum_{j=1}^{N} \mu_j \frac{\partial H_{t+1}}{\partial e_i} - \frac{\partial C_{\text{policy}}}{\partial e_i}. \tag{7}$$

The private condition contains $b_i$ and $\mu_i$, while the social condition contains the aggregate benefit and aggregate pressure exposure. This externality is the mathematical source of under-contribution in the decentralized benchmark.

**Proposition 2** (Local marginal source of free riding)**.** At an interior baseline point away from projection boundaries and threshold discontinuities, the decentralized marginal incentive omits the positive spillover from agent $i$'s effort to the other agents. Therefore, when the remaining agents have positive aggregate benefit or pressure exposure, the social marginal value of effort is larger than the private marginal value before policy correction.

*Proof.* For an interior baseline point, Equations 1 and 3 imply

$$\frac{\partial \widetilde{G}_{t+1}}{\partial e_i} = \alpha (1 + m) \Gamma_G(G_t) \frac{a_i}{(1 + Q_t) \log(1 + Q_{\max})} \geq 0,$$
$$\frac{\partial \widetilde{H}_{t+1}}{\partial e_i} = -\kappa \frac{a_i}{(1 + Q_t) \log(1 + Q_{\max})} < 0.$$

Subtracting the private marginal condition in Equation 6 from the social marginal condition in Equation 7 gives

$$\frac{\partial W}{\partial e_i} - \frac{\partial u_i}{\partial e_i} = \left(\sum_{j\neq i} b_j\right) V'(G_{t+1}) \frac{\partial \widetilde{G}_{t+1}}{\partial e_i} - \left(\sum_{j\neq i} \mu_j\right) \frac{\partial \widetilde{H}_{t+1}}{\partial e_i} - \frac{\partial C_{\text{policy}}}{\partial e_i} - \text{private policy terms}.$$

For the baseline comparison, the direct policy terms vanish. The first spillover term is nonnegative because increasing $e_i$ raises next-period public-good stock for other beneficiaries unless the capacity boundary is already binding. The second spillover term is positive because $\partial \widetilde{H}_{t+1}/\partial e_i < 0$, so the same effort reduces pressure exposure for other agents. Thus the private response ignores positive marginal effects on others, which explains why the individual-rational response tends to stop before the social-planner benchmark under the synthetic baseline settings. □

## 5.2 Damped Marginal-Response Update

The decentralized solution is obtained by damped marginal-response updates. At each iteration, an agent responds to marginal public benefit, marginal pressure relief, private cost, and policy incentives. To match the numerical calculation, we write the raw marginal response as

$$\widetilde{MR}_i^{(k)} = \frac{1}{c_i}\Bigg[0.90\, a_i b_i \alpha(1+m)\Gamma_G(G_t)V'(G_t)q'(Q^{(k)}) + 0.85\, a_i\mu_i\kappa q'(Q^{(k)}) + s + \frac{r}{1+e_i^{(k)}} + p\left(0.35 + 2.8(\tau^\star - e_i^{(k)})_+\right)\mathbf{1}_{\{b_i\geq \bar{b}_q\}} + 0.10p\,\mathbf{1}_{\{e_i^{(k)}<\tau^\star\}}\Bigg],$$

where $q'(Q) = \{(1+Q)\log(1+Q_{\max})\}^{-1}$, $\Gamma_G(G_t)$ is the stock saturation factor, and $\tau^\star = \max(\tau, \epsilon)$ is the implemented penalty threshold. The constants 0.90, 0.85, 0.35, 2.8, and 0.10 are synthetic behavioral-response weights used in the numerical calculation to obtain a stable marginal-response proxy rather than a closed-form analytical Nash equilibrium. They are not empirical estimates; they rescale the marginal terms in the computational response rule, and the sensitivity, robustness, and Monte Carlo analyses below are used to check that the qualitative conclusions are not tied to one exact response calibration. The projected and damped update is

$$e_i^{(k+1)} = 0.58 e_i^{(k)} + 0.42\Pi_{[0,e_{\max}]}\left(\widetilde{MR}_i^{(k)}\right),$$

where $\Pi$ denotes projection onto the feasible interval. This is why the paper refers to the output as a Nash-style individual-rational solution instead of a proven unique Nash equilibrium.

**Table 4:** Computational procedure for the Nash-style individual-rational solution.

| Step | Operation | Description |
|---|---|---|
| 1 | Initialize | Set a feasible initial effort vector $\mathbf{e}^{(0)} \in [0, e_{\max}]^N$. |
| 2 | Individual update | For each agent $i$, compute a marginal response $MR_i$ from the private first-order balance and project it to $[0, e_{\max}]$. |
| 3 | Damping | Update $\mathbf{e}^{(k+1)} = 0.58\mathbf{e}^{(k)} + 0.42\Pi(BR(\mathbf{e}^{(k)}))$. |
| 4 | Stopping rule | Stop when $\max_i \lvert e_i^{(k+1)} - e_i^{(k)}\rvert$ is below the tolerance or when the maximum iteration number is reached. |

In the numerical calculation, the initial effort is 0.06 for every agent, the damping weight on the raw projected response is 0.42, the tolerance is $10^{-7}$, and the maximum number of sweeps is 8. Because the maximum sweep limit is reached before exact convergence in the reported baseline checks, we call the resulting effort vector an approximate Nash-style individual-rational solution. This wording is intentional: we use a numerical equilibrium proxy and do not claim uniqueness of an analytical Nash equilibrium. The numerical check in Table 5 shows that the damped update is stable in the four baseline scenarios; the remaining raw marginal-response residual is small relative to the feasible effort interval $[0, 1]$.

**Table 5:** Numerical stability check for the baseline Nash-style marginal-response update.

| Scenario | Damped sweeps | Max residual | $Q^{IR}$ | Mean effort |
|---|---|---|---|---|
| Small volunteer | 8 | 0.0017 | 2.688 | 0.074 |
| Rapid growth | 8 | 0.0030 | 3.344 | 0.100 |
| Critical infrastructure | 8 | 0.0016 | 3.007 | 0.089 |
| Burnout-prone | 8 | 0.0016 | 1.601 | 0.046 |

*The residual is reported after the fixed eight damped sweeps used in the reported simulation setting. It is a diagnostic for a stable computational proxy, not a proof of exact convergence or uniqueness of a Nash equilibrium.*

## 5.3 Free-Rider Classification

An agent is classified as a free rider if it has high benefit but contributes below a low-effort threshold. Let $\bar{b}_q$ be the high-benefit quantile cutoff and let $\epsilon$ be the free-rider effort threshold. Then

$$F_i = \mathbf{1}\{e_i < \epsilon,\ b_i \geq \bar{b}_q\}.$$

The free-riding ratio is

$$FR = \frac{1}{N}\sum_{i=1}^{N} F_i.$$

In the reported simulation setting, $\epsilon = 0.08$ and $q = 0.55$, so an agent must be both below the low-effort threshold and above the 55th percentile of the benefit distribution to be classified as a free rider. This definition avoids labeling low-benefit agents as free riders simply because they contribute little. It also makes the indicator conservative: passive agents with little expected benefit are not counted as free riders unless they are among the higher-benefit participants.

For policy evaluation, this conservative definition is preferable to a pure low-effort rule. A low-effort rule would classify almost every capacity-constrained or low-benefit participant as a governance failure, even when their private gain from the public good is limited. By adding the high-benefit condition, the indicator focuses on agents who have a stronger reason to enjoy the public good but still choose a low contribution level. This makes the free-riding ratio more closely connected to the strategic incentive problem studied in the paper.

Figure 3 provides a visual check of this definition under the synthetic agent population. The distribution is not used to estimate real behavior; it is used to show that the classification rule separates low-effort, high-benefit agents from low-effort agents with limited private benefit. This distinction is important because the policy mechanisms target strategic free riding rather than merely low contribution.

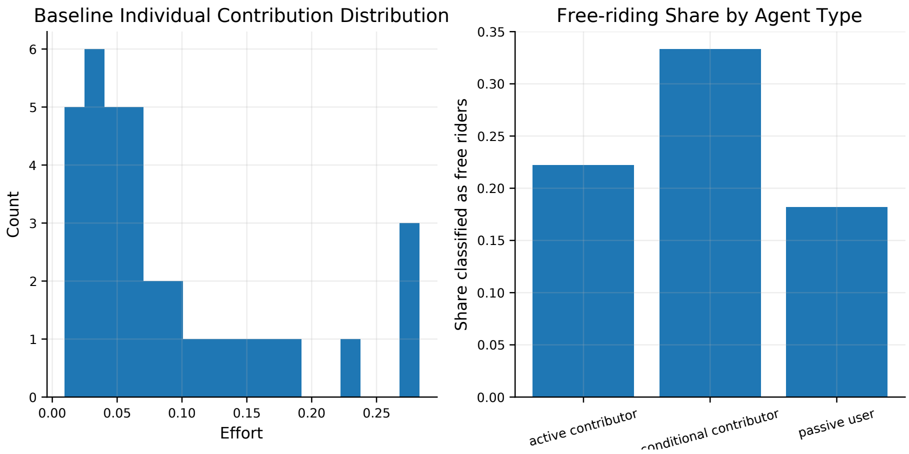


**Figure 3:** Synthetic contribution and benefit distribution used for free-rider classification. The figure illustrates how the model distinguishes high-benefit low-effort free riders from low-benefit passive agents.

# 6 Model III: Social-Planner Benchmark and Welfare Loss

## 6.1 Stage-Wise Social-Planner Benchmark

The social planner chooses the vector of efforts to maximize total welfare after accounting for policy cost:

$$W(\mathbf{e}_t) = \sum_{i=1}^{N} \left[ b_i V(G_{t+1}) - \frac{1}{2} c_i e_{i,t}^2 - \mu_i H_{t+1} \right] - C_{\text{policy}}.$$

The stage-wise social-planner benchmark is

$$\mathbf{e}_t^{SO} = \arg \max_{\mathbf{e}_t \in [0, e_{\max}]^N} W(\mathbf{e}_t).$$

**Proposition 3** (Existence of the stage-wise planner benchmark)**.** For every fixed synthetic scenario state $(G_t, H_t, D_t)$, the baseline stage-wise planner problem used in the individual-rational versus social-planner comparison has at least one feasible maximizer $\mathbf{e}_t^{SO} \in [0, e_{\max}]^N$.

*Proof.* The feasible effort set $[0, e_{\max}]^N$ is closed and bounded, hence compact. For a fixed state and fixed agent attributes, the aggregate contribution $Q_t = \sum_i a_i e_{i,t}$ and the normalized effectiveness $q(Q_t)$ are continuous in $\mathbf{e}_t$. The unprojected stock and pressure transitions are therefore continuous functions of $\mathbf{e}_t$, and projection onto closed intervals preserves continuity. Since $V(G) = \log(1 + \omega G)$, the benefit term, quadratic effort cost, and pressure term in the baseline welfare function are continuous in $\mathbf{e}_t$. By the extreme value theorem, the continuous welfare function attains a maximum on the compact feasible set. Therefore a stage-wise planner maximizer exists. This proposition only establishes existence of a finite-dimensional benchmark; it does not imply uniqueness or solve an infinite-horizon social optimum. □

The optimization is solved numerically using L-BFGS-B with box constraints, maximum iteration number 120, and function tolerance $10^{-10}$. This benchmark is a tractable upper reference for each current state rather than a full infinite-horizon dynamic-programming solution. This choice is appropriate for the present task because the main objective is to diagnose the mechanism and magnitude of decentralized under-provision under comparable state conditions. Numerical checks show that the stage-wise social-planner benchmark is valid in all four scenarios, with higher contribution and higher welfare than the Nash-style solution. Proposition 3 justifies that the finite-dimensional baseline benchmark is well-defined; we do not claim uniqueness, nor do we claim to solve an infinite-horizon social optimum.

## 6.2 Individual-Rational and Social-Planner Comparison

Table 6 and Figure 4 compare the individual-rational benchmark and the social-planner benchmark under the updated capacity-saturating dynamics. Here the superscript $IR$ denotes the Nash-style individual-rational solution. Under the synthetic scenario settings, the Nash-style aggregate effective contribution is much lower than the social-planner contribution in every scenario. This is exactly the free-riding mechanism described in the problem and in classical analyses of collective action: individual agents internalize their private cost but only partially internalize the shared benefit of contribution [2], [3].

**Table 6:** Nash-style individual rationality versus stage-wise social-planner benchmark under synthetic scenarios.

| Scenario | $Q^{IR}$ | $Q^{SO}$ | $G^{IR}$ | $G^{SO}$ | Free-riding gap | Under-provision | Welfare loss |
|---|---|---|---|---|---|---|---|
| Small volunteer | 2.688 | 14.069 | 0.743 | 0.966 | 0.809 | 0.231 | 0.101 |
| Rapid growth | 3.344 | 17.149 | 0.728 | 0.940 | 0.805 | 0.226 | 0.116 |
| Critical infrastructure | 3.007 | 17.453 | 0.737 | 1.013 | 0.828 | 0.272 | 0.147 |
| Burnout-prone | 1.601 | 11.455 | 0.488 | 0.695 | 0.860 | 0.299 | 0.198 |

*The benchmark is stage-wise and simulation-based. It is used to diagnose under-provision under identical state conditions, not to claim an infinite-horizon global optimum.*

Table 6 reports one-step benchmark outcomes under an identical current state. It is therefore used for mechanism diagnosis rather than for long-run policy ranking. Later policy tables report tail averages from dynamic trajectories, so their welfare numbers should not be compared mechanically with the one-step values in this benchmark table. The qualitative conclusion is unchanged after introducing capacity saturation: decentralized contribution remains far below the social-planner benchmark in all four scenarios.

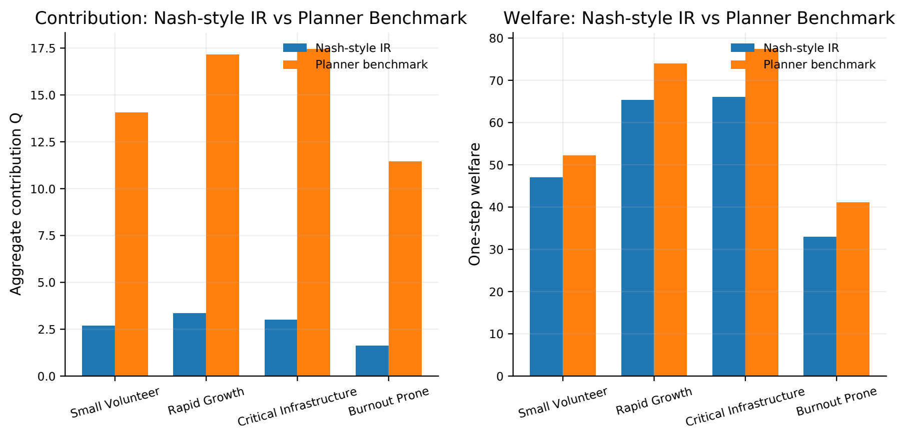


**Figure 4:** Nash-style individual rationality versus social-planner benchmark. Controlled numerical experiments show lower contribution and welfare under decentralized individual rationality.

Table 6 and Figure 4 should be read together. The table provides the numerical gap, while the figure makes the direction of the gap visually immediate. In all four scenarios, the social-planner benchmark selects a much higher aggregate effective contribution than the Nash-style individual-rational solution. This is consistent with the classical public-good mechanism: each individual receives only a fraction of the social benefit from its own contribution but bears the full private cost. Therefore, decentralized contribution is lower than socially coordinated contribution.

The welfare gap is not proportional only to the contribution gap. For example, the burnout-prone scenario has the highest free-riding gap and the highest welfare loss, but the size of welfare loss also depends on cost and pressure dynamics. This distinction matters because a policy that increases contribution may not be desirable if it imposes excessive cost or fails to reduce pressure. For this reason, the later policy section evaluates interventions by welfare, cost, free-riding reduction, pressure, fairness, and stability rather than by contribution alone.

Figure 5 extends the comparison from a one-step benchmark to a dynamic trajectory. It shows that the social-planner benchmark does not merely increase the initial contribution level; it changes the path of stock accumulation and pressure relief over time. This dynamic view is closer to the public-good problem statement, because free riding becomes costly through repeated under-maintenance rather than through a single isolated decision.

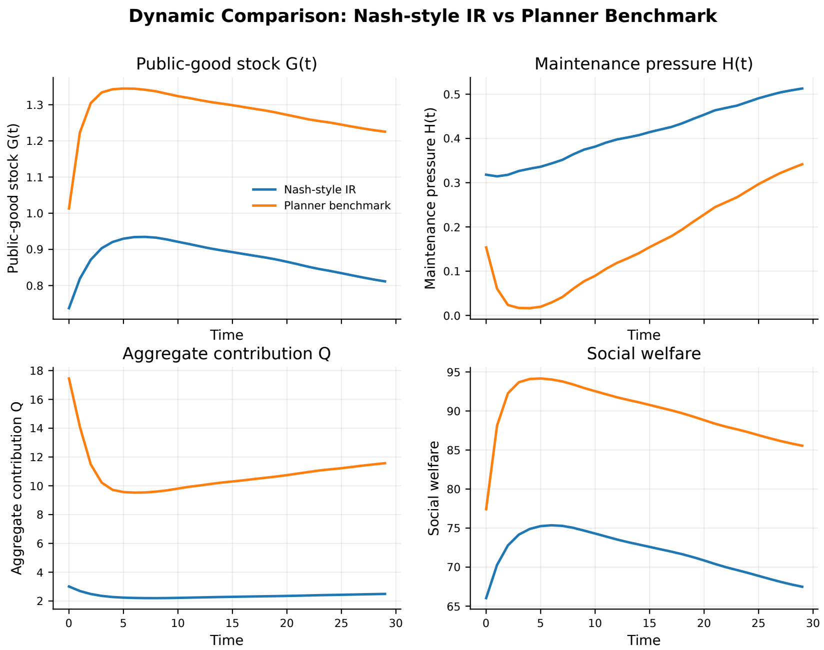


**Figure 5:** Dynamic trajectory comparison between Nash-style individual rationality and the stage-wise social-planner benchmark. The trajectory view shows how under-provision accumulates through repeated contribution decisions.

### 6.3 Free-Riding Gap, Under-Provision, and Welfare Loss

We define the three main diagnostic measures as

$$\mathrm{FreeRidingGap} = \frac{Q^{SO} - Q^{IR}}{Q^{SO}},$$

$$\mathrm{UnderProvision} = \frac{G^{SO} - G^{IR}}{G^{SO}},$$

$$\mathrm{WelfareLoss} = \frac{W^{SO} - W^{IR}}{|W^{SO}|}.$$

The synthetic scenarios produce a free-riding gap between 0.805 and 0.860, an under-provision ratio between 0.226 and 0.299, and a welfare loss ratio between 0.101 and 0.198. Figure 6 decomposes the gap by scenario and shows that the burnout-prone scenario has the largest welfare loss.

The three ratios should not be interpreted as interchangeable. The free-riding gap is closest to the decision mechanism because it compares contribution levels. The under-provision ratio translates this decision gap into stock shortage. The welfare-loss ratio is the final integrated consequence after benefits, costs, and pressure terms are combined. Therefore, a scenario can have a large free-riding gap but a smaller welfare loss if the stock system absorbs part of the shortage; conversely, high pressure can turn a comparable contribution gap into a larger welfare loss. This distinction is the reason we report the decomposition rather than only one aggregate score.

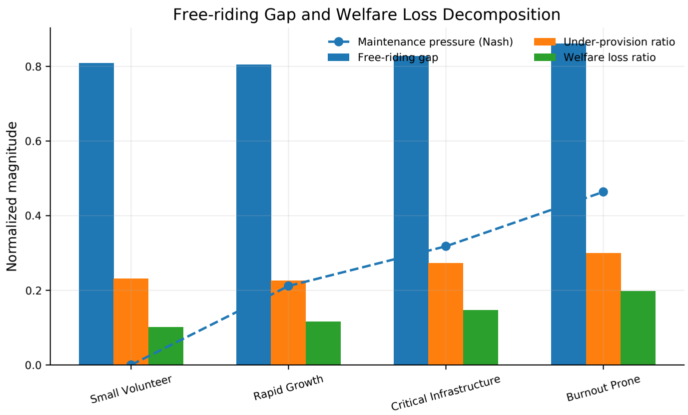


**Figure 6:** Welfare-loss decomposition under synthetic scenarios. The figure connects the free-riding gap, under-provision ratio, welfare loss ratio, and maintenance pressure index.

Figure 6 separates four related but different outcomes. The free-riding gap measures the missing effective contribution relative to the planner benchmark. The under-provision ratio measures the resulting shortage in public-good stock. The welfare-loss ratio measures the final utility loss after benefit, cost, and pressure are combined. Maintenance pressure records the dynamic burden carried by the system. Presenting all four indicators prevents the analysis from reducing free riding to only one number.

The decomposition also suggests why scenario design matters. The burnout-prone scenario has the most severe welfare loss because high cost discourages effort and high pressure magnifies the loss from inadequate maintenance. The critical-infrastructure scenario has a smaller welfare-loss ratio than burnout-prone, but it has high pressure and is therefore a natural testbed for policy intervention. These patterns are simulation-based, but they provide a clear mechanism-oriented interpretation: the social cost of free riding is amplified when contribution is expensive and when pressure has strong negative feedback on the public-good stock.

# 7 Model IV: Policy Intervention Mechanisms

## 7.1 Policy Variables

We study six intervention mechanisms in addition to the baseline. The policy set is designed to cover both material incentives and behavioral incentives, reflecting the broader literature on cooperation, punishment, reputation, and warm-glow contribution [8], [9], [10]:

- **Subsidy** reduces the private net cost of contribution.
- **Penalty** discourages high-benefit agents from contributing below a threshold.
- **Reputation** gives additional concave utility to contribution.
- **Matching fund** amplifies the effect of contribution on public-good stock.
- **Threshold governance** combines a minimum contribution threshold with budget and backlog reduction.
- **Combined portfolio** mixes subsidy, penalty, reputation, matching, budget, and backlog reduction.

Table 7 reports the policy intensities used in the deterministic comparison.

**Table 7:** Synthetic policy settings used in the deterministic policy comparison.

| Policy | $s$ | $p$ | $r$ | $m$ | $B$ | $R$ | $\tau$ |
|---|---|---|---|---|---|---|---|
| Baseline | 0.00 | 0.00 | 0.00 | 0.00 | 0.000 | 0.00 | 0.00 |
| Subsidy | 0.20 | 0.00 | 0.00 | 0.00 | 0.000 | 0.00 | 0.00 |
| Penalty | 0.00 | 0.35 | 0.00 | 0.00 | 0.000 | 0.01 | 0.16 |
| Reputation | 0.00 | 0.00 | 0.26 | 0.00 | 0.000 | 0.00 | 0.00 |
| Matching fund | 0.00 | 0.00 | 0.00 | 0.30 | 0.000 | 0.00 | 0.00 |
| Threshold governance | 0.00 | 0.18 | 0.00 | 0.00 | 0.025 | 0.05 | 0.18 |
| Combined portfolio | 0.12 | 0.14 | 0.16 | 0.18 | 0.020 | 0.04 | 0.14 |

*These values are not empirical estimates; they are controlled settings chosen to compare pure and mixed intervention mechanisms under the same dynamic environment.*

## 7.2 Policy Cost Function

Policy cost is explicitly included:

$$C_{\text{policy}} = s\sum_i e_i + 0.35m\sum_i e_i + 0.20r\log\left(1+\sum_i e_i\right) + 50B + 35R + 0.03pN_F,$$

where $N_F$ is the number of classified free riders. The coefficients are synthetic governance-cost weights. They are used to prevent a policy from appearing attractive merely because its administrative cost is ignored.

## 7.3 Marginal Effects of Policy Instruments

The policy terms change the individual marginal condition in different ways. For an interior effort level and away from the non-differentiable threshold point, subsidy contributes

$$\frac{\partial(se_i)}{\partial e_i} = s,$$

so it directly raises the private marginal return to effort. Reputation contributes

$$\frac{\partial r\log(1+e_i)}{\partial e_i} = \frac{r}{1+e_i},$$

which is positive but decreasing; it is therefore most effective for moving very low contributors away from zero. For an agent below the contribution threshold, the implemented penalty incentive is not a single quadratic term. It follows the same marginal-response rule used in the Nash-style update:

$$p\left(0.35 + 2.8(\tau^\star - e_i)_+\right)\mathbf{1}_{\{b_i \geq \bar{b}_q\}} + 0.10p\mathbf{1}_{\{e_i < \tau^\star\}}.$$

The first component targets high-benefit low contributors, while the second gives all below-threshold agents a small upward push. Matching fund changes the production side by multiplying $\partial G_{t+1}/\partial e_i$ by $(1+m)$, so it increases the stock productivity of contribution rather than directly punishing free riding. Budget injection $B$ shifts the stock equation, while backlog reduction $R$ shifts the pressure equation. Thus different policies act on different parts of the marginal-incentive decomposition.

## 7.4 Multi-Criteria Evaluation

A policy is evaluated by a vector of indicators:

$$\mathcal{E}(\pi) = \left(\overline{W}, \overline{C}_{\text{policy}}, \Delta FR, \Delta H, \text{Gini}(e), \text{Stability}\right).$$

The tail averages are computed over the long-run part of each trajectory. In the numerical calculation this is the last $\max(8, \lfloor T/5 \rfloor)$ periods; under the reported simulation setting with $T = 30$, it is therefore the last eight periods. Effort inequality is measured by the Gini index

$$\text{Gini}(e) = \frac{2\sum_{i=1}^{N} i e_{(i)}}{N\sum_{i=1}^{N} e_{(i)}} - \frac{N+1}{N},$$

where $e_{(i)}$ is the sorted effort vector. Local dynamic stability is summarized by

$$\text{Stability} = 1 - \rho(J_t),$$

where $J_t$ is the local Jacobian of the state-transition map and $\rho(J_t)$ is its spectral radius. For this local stability diagnostic, the Jacobian is computed analytically from the unclipped saturating transition and interpreted as a local diagnostic away from projection boundaries. We use this multi-criteria approach because policy recommendations should not depend only on welfare. Cost, pressure, fairness, and stability are also relevant for governance.

# 8 Model Verification and Numerical Reliability

Before interpreting policy outcomes, we check whether the numerical framework is internally consistent. Table 8 summarizes the main reliability checks used in the paper. These checks do not turn the synthetic scenarios into empirical evidence; rather, they show that the reported conclusions follow from a reproducible and stable computational design. The paper reports only the tables and figures that support distinct modeling conclusions, while numerical values are rounded to displayed precision. Independent reruns may produce machine-precision differences in optimization-based quantities, but the reported rankings and qualitative conclusions are expected to remain unchanged at the displayed rounding level.

**Table 8:** Model checking and numerical reliability evidence.

| Check | Evidence | Interpretation |
|---|---|---|
| Reproducible synthetic route | Fixed seed 20260614; four scenarios; seven policies; all inputs and outputs are treated as controlled synthetic simulations. | Results can be regenerated and interpreted as controlled simulations. |
| Best-response stability | The maximum residual in the baseline scenarios is at most 0.0030 over the feasible interval $[0, 1]$. | The individual-rational benchmark is numerically stable under the tested settings. |
| Planner benchmark validity | The stage-wise L-BFGS-B solution gives higher contribution and welfare than the decentralized benchmark in all four scenarios. | The planner benchmark is a consistent upper reference for diagnosing under-provision. |
| Scenario separation | Small volunteer, rapid growth, critical infrastructure, and burnout-prone settings stress different cost, pressure, and demand mechanisms. | The scenario analysis is not a repetition of one parameter setting. |
| Uncertainty tests | One-dimensional sensitivity, two-dimensional response surface, perturbed-scenario robustness, Monte Carlo runs, and Pareto search are all reported. | The qualitative conclusions are tested beyond one baseline parameter vector. |
| Displayed-precision reproducibility | The numerical design records the fixed random seed and synthetic settings; optimization results are interpreted at table-rounding precision. | Small floating-point differences do not change the reported ranking or qualitative conclusion. |

# 9 Numerical Experiments and Results

## 9.1 Baseline Dynamic Behavior

Figure 7 summarizes stock, pressure, free-riding, and welfare trajectories under baseline governance. The dashboard is placed in the result section rather than in the model-construction section because it is used as numerical evidence from the synthetic simulation, not as part of the mathematical definition.

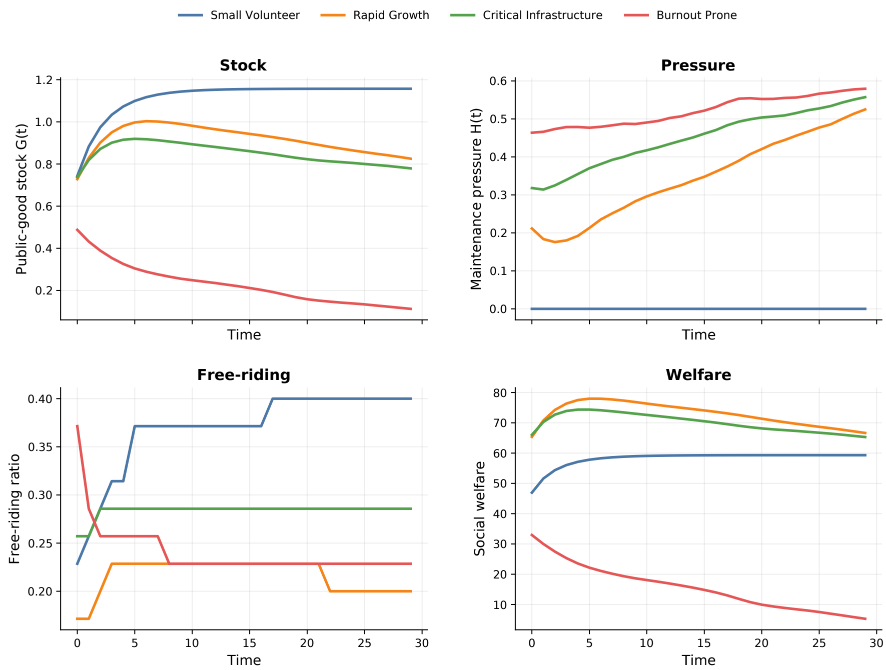


**Figure 7:** Baseline dynamic behavior under synthetic scenarios. The panels compare stock, maintenance pressure, free-riding ratio, and welfare under the baseline Nash-style simulation.

The upper panels of Figure 7 indicate that public-good stock and maintenance pressure are coupled but not identical. A scenario can maintain a moderate stock for several periods while pressure is still building in the background. The lower panels show that free riding and welfare are also scenario-dependent. Thus, free-riding behavior is not only an individual-level phenomenon; through stock and pressure feedback, it becomes a dynamic system-level problem.

The baseline simulations indicate that free riding and under-provision are not uniform across scenarios. Small-volunteer provision reaches a relatively stable stock with almost no maintenance pressure, but still exhibits a nonzero free-riding ratio. This means that a system can appear operational even when the contribution burden is unevenly distributed. Rapid growth and critical infrastructure display higher maintenance pressure, showing that demand growth and infrastructure importance make the public-good problem more dynamic. The burnout-prone scenario shows the most severe welfare loss when compared with the social-planner benchmark.

These baseline results serve two roles in the paper. First, they verify that the synthetic scenarios are not redundant: each scenario stresses a different component of the mechanism. Second, they justify selecting the critical-infrastructure scenario for detailed policy comparison. This scenario is neither trivially stable nor completely collapsed; it has enough free riding, pressure, and welfare loss for policy differences to be visible. The results should be interpreted as synthetic numerical patterns rather than real-world measurements.

Figure 8 gives a three-dimensional phase view of representative trajectories in the stock-pressure-welfare space. This figure is not an additional metric; it is a geometric summary of the dynamic mechanism. The baseline path remains in a lower-welfare region with higher pressure, while reputation and combined portfolio policies move the trajectory toward higher stock and welfare. The separation of these paths supports the interpretation that policy effects operate through the coupled stock-pressure system, not only through an immediate change in effort.

This phase-space interpretation is useful because two trajectories can have similar average welfare for a short period while having different pressure accumulation patterns. A path with high pressure is more fragile: once demand or decay is perturbed, the same contribution level may no longer maintain the stock. Therefore, the trajectory figure links the baseline dashboard to the policy section by showing why the critical-infrastructure scenario should be evaluated by both welfare and dynamic resilience.

The main modeling message is that public-good failure is not only a low-contribution event. It is a state-space drift: weak private incentives reduce contribution, insufficient contribution raises pressure, and pressure then lowers the future welfare effect of the same governance system. The figure is therefore placed before the policy-comparison table so that the reader first sees the dynamic failure channel and then evaluates which interventions move the system away from it.

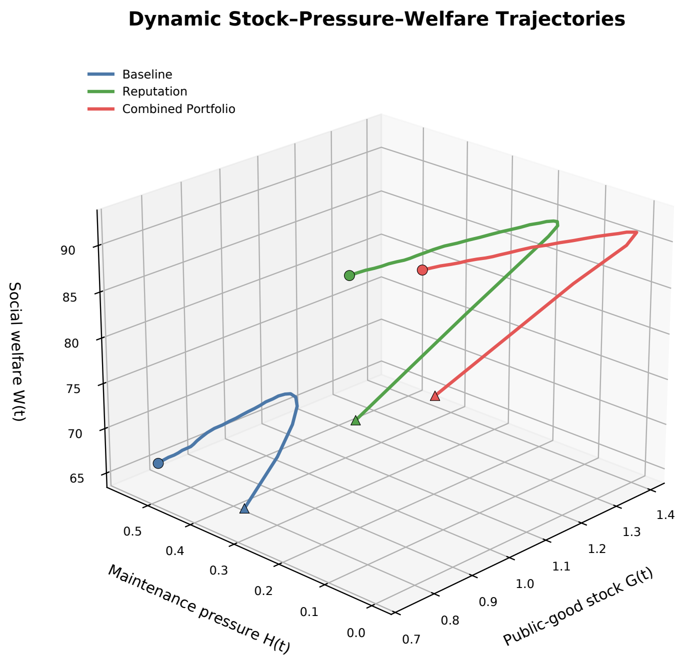


**Figure 8:** Three-dimensional phase trajectory in the stock-pressure-welfare space. The figure visualizes how baseline governance, reputation, and combined portfolio policies move through different dynamic regions.

The phase trajectory also gives a compact explanation of why welfare alone is not enough. The baseline path first moves to a relatively high stock but then drifts toward higher pressure and lower welfare, which means that apparent short-run service quality can hide accumulated maintenance burden. Reputation shifts the trajectory upward with limited cost because it changes marginal incentives without directly adding a large budget term. The combined portfolio follows a different path: it does not simply maximize welfare at each moment, but it pulls the system toward a lower-pressure region. This distinction is important for public-good governance, because a policy that is slightly weaker in welfare may still be preferred if it reduces the probability of a future pressure-driven collapse.

## 9.2 Policy Comparison in the Critical-Infrastructure Scenario

The critical-infrastructure scenario is selected for the main policy comparison because it combines high public-good importance, visible free riding, and nontrivial maintenance pressure. Table 9 and Figure 9 show that subsidy, penalty, reputation, threshold governance, and the combined portfolio all improve welfare relative to the baseline. Under the updated capacity-saturating dynamics, the combined portfolio has the highest long-run welfare and the largest maintenance-pressure reduction, but it also has the highest policy cost. Reputation produces nearly the same welfare level with a much smaller cost, so it is the strongest low-cost behavioral instrument.

**Table 9:** Policy comparison in the critical-infrastructure synthetic scenario.

| Policy | Welfare | Cost | Free riding | Pressure | $\Delta FR$ | $\Delta H$ | Effort Gini |
|---|---|---|---|---|---|---|---|
| Baseline | 66.52 | 0.00 | 0.286 | 0.533 | 0.000 | 0.000 | 0.446 |
| Subsidy | 82.13 | 1.89 | 0.057 | 0.389 | 0.229 | 0.144 | 0.410 |
| Penalty | 76.68 | 0.35 | 0.000 | 0.468 | 0.286 | 0.065 | 0.393 |
| Reputation | 83.74 | 0.12 | 0.000 | 0.392 | 0.286 | 0.141 | 0.358 |
| Matching fund | 72.77 | 0.27 | 0.286 | 0.549 | 0.000 | -0.017 | 0.446 |
| Threshold governance | 75.61 | 3.02 | 0.089 | 0.443 | 0.196 | 0.090 | 0.397 |
| Combined portfolio | 83.81 | 4.58 | 0.000 | 0.311 | 0.286 | 0.221 | 0.360 |

*Tail averages over the last eight periods are used for long-run indicators. $\Delta FR$ and $\Delta H$ are reductions relative to the baseline; a negative $\Delta H$ means pressure is higher than baseline.*

Unlike Table 6, Table 9 uses long-run tail averages over the last eight periods. The comparison therefore measures sustained policy performance rather than a one-period response gap. This distinction explains why the baseline welfare level in the policy table is not identical to the one-step individual-rational benchmark.

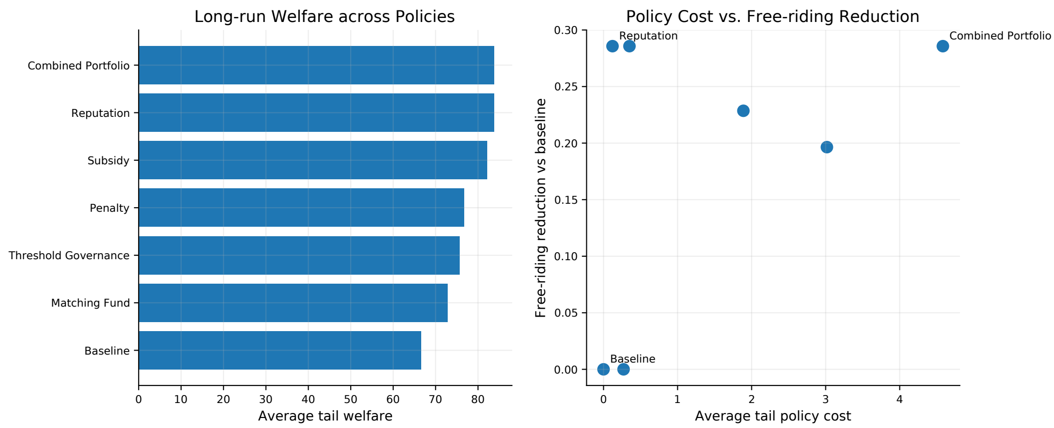


**Figure 9:** Policy comparison under the critical-infrastructure synthetic scenario. Interventions improve welfare and reduce free riding, but differ in policy cost and pressure reduction.

The comparison shows three policy patterns. First, penalty, reputation, and the combined portfolio eliminate classified free riding in the critical-infrastructure setting. Their mechanisms are different: penalty changes the downside of under-contribution for high-benefit agents, while reputation creates a positive private return to contribution. Second, matching fund improves welfare relative to the baseline but does not reduce the classified free-riding ratio and does not provide pressure relief under the current parameterization. This happens because matching mainly raises the productivity of whatever contribution already occurs; it does not directly change the classification of high-benefit low-effort agents. Third, the combined portfolio is strongest in pressure reduction, but it has the highest policy cost because it simultaneously changes private incentives and the stock-pressure state. Thus, the best policy depends on the objective: reputation is attractive when cost effectiveness is central, while the combined portfolio is more attractive when maintenance pressure is the dominant risk.

The policy comparison also explains why the paper avoids a single "optimal policy" claim. In the simulation, combined portfolio has the highest welfare and pressure reduction, reputation has almost the same welfare and much lower cost, and penalty has the strongest low-cost free-riding correction. These rankings are not identical across criteria. This is typical for public-good governance: incentive correction, administrative cost, fairness, and system stability cannot usually be optimized simultaneously by one instrument.

## 9.3 Multi-Dimensional Policy Ranking

Figure 10 compares selected policies by multiple normalized dimensions. The radar plot is not used as proof of a single best policy. Instead, it shows that governance objectives are plural. A policy can be strong in welfare and free-riding reduction but weaker in cost, fairness, or pressure relief. Therefore, the model suggests a scenario-specific portfolio rather than a universal policy.

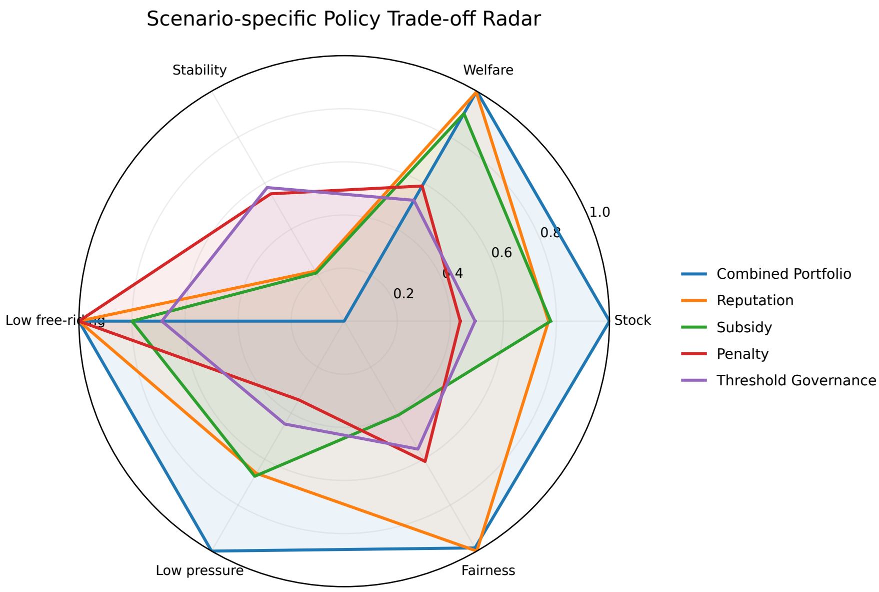


**Figure 10:** Multi-dimensional policy comparison under synthetic settings. All axes are normalized for visualization only; the radar area is not used as a formal optimization score.

The radar chart is included to communicate the multi-objective nature of the policy problem. It should not be over-interpreted as an exact optimization result, because normalization changes visual distances. Its value is comparative: it shows that a policy with a large area on one set of criteria may still be weak on another. For example, a low-cost policy may be welfare-effective but not necessarily the strongest pressure-control policy. Conversely, a portfolio policy may improve system protection at the expense of higher intervention cost.

For the present analysis, this visual form is useful because the mathematical model contains both agent-level and system-level indicators. Free-riding reduction is an agent-behavior metric, maintenance pressure is a system-burden metric, and welfare combines benefit and cost. A policy recommendation based on only one of them would ignore part of the public-good problem.

## 9.4 Pareto Trade-Off Analysis

**Definition 1** (Policy dominance)**.** For two policy portfolios $\pi_a$ and $\pi_b$, policy $\pi_a$ dominates $\pi_b$ if it is no worse in all six implemented indicators,

$$\overline{W}(\pi_a) \geq \overline{W}(\pi_b), \qquad \overline{G}(\pi_a) \geq \overline{G}(\pi_b), \qquad \overline{C}_{\text{policy}}(\pi_a) \leq \overline{C}_{\text{policy}}(\pi_b),$$
$$\overline{FR}(\pi_a) \leq \overline{FR}(\pi_b), \qquad \overline{H}(\pi_a) \leq \overline{H}(\pi_b), \qquad \text{Gini}(\pi_a) \leq \text{Gini}(\pi_b),$$

and at least one of these inequalities is strict. A policy is Pareto-efficient if no sampled policy dominates it.

A random policy search is used to examine the trade-off among welfare, stock, cost, free-riding reduction, pressure, and effort inequality. Figure 11 shows a two-dimensional projection of this six-indicator dominance test: dominated policies are those for which another sampled portfolio is no worse in all six indicators and strictly better in at least one. Pareto-efficient policies usually combine modest penalty, reputation, and selective backlog reduction. The important conclusion is not that one exact parameter vector is optimal, but that policy design has a frontier structure.

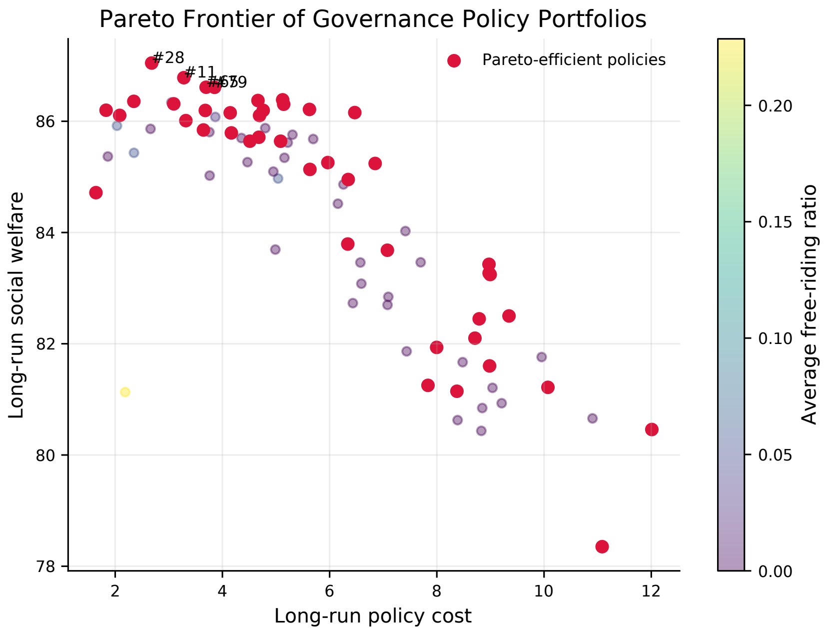


**Figure 11:** Pareto frontier for synthetic policy trade-offs. The plot shows welfare against policy cost, with color indicating free-riding ratio; highlighted points are Pareto-efficient under the full six-indicator dominance rule used in the numerical calculation.

Figure 11 provides the most direct trade-off view, but it should be read as a projection of the full dominance calculation rather than as a purely two-dimensional frontier. A dominated point means that another sampled policy portfolio achieves at least as much welfare and stock, no greater cost, no worse free-riding, no higher pressure, and no higher effort inequality, with at least one strict improvement. The frontier points represent policies that remain competitive after these comparisons. For policy design, this is more informative than reporting a single maximum-welfare setting, because real decision makers often face budget limits and institutional constraints.

Figure 12 adds a three-dimensional projection of the same sampled policy search. The vertical and depth coordinates help reveal a pattern that can be hidden in the two-dimensional cost-welfare plot: some portfolios with similar welfare and cost can still differ in pressure control and effort inequality. Therefore, the Pareto analysis is interpreted as a multi-criteria selection problem rather than as a one-dimensional ranking.

The scatter pattern also explains why the paper does not report a single policy parameter as “the optimum”. If governance capacity is limited, a decision maker may prefer a point with slightly lower welfare but lower cost and lower inequality. If system reliability is the priority, a point with larger intervention cost may be justified because it reduces maintenance pressure. The sampled frontier is therefore used as a decision map rather than as a deterministic prescription.

From a modeling perspective, the Pareto view is also a guard against a common reporting error: choosing the largest welfare value and ignoring its side effects. In the sampled set, some high-welfare portfolios are dominated once cost, free riding, pressure, stock, and effort inequality are considered together. This supports the final recommendation that public-good policy should be selected by trade-off position, not by a single scalar score.

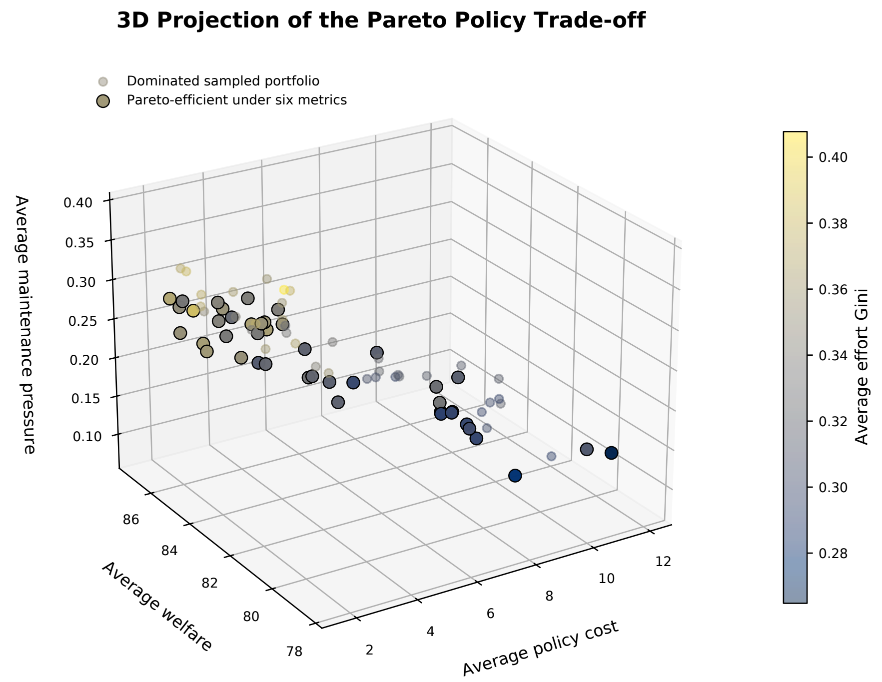


**Figure 12:** Three-dimensional projection of the synthetic Pareto policy search. Highlighted portfolios remain Pareto-efficient under the full six-indicator dominance rule.

The three-dimensional projection further shows that Pareto-efficient points are not concentrated at the lowest possible policy cost or at the maximum welfare corner. Several efficient portfolios occupy a middle region where cost is moderate, welfare remains high, and effort inequality is not excessive. This pattern supports a governance interpretation: once free riding has been largely corrected, additional spending may generate smaller welfare gains and can even worsen the fairness-cost balance. Thus, the frontier should be read as a screening tool for feasible policy portfolios rather than as a command to choose the most expensive or most punitive setting.

The Pareto analysis also prevents over-fitting the policy recommendation to one named policy. Named policies such as reputation or penalty are interpretable mechanisms, but actual governance may combine them with different intensities. The frontier experiment shows that mixed portfolios can be competitive, especially when they combine incentive correction with moderate pressure relief. Therefore, the policy section recommends principles and scenario-specific portfolios rather than exact numerical rates.

# 10 Sensitivity, Robustness, and Uncertainty Analysis

## 10.1 One-Dimensional Sensitivity

To test whether the conclusions depend on a single parameter choice, we vary key model and policy parameters in the critical-infrastructure scenario. Table 10 lists the parameters with the largest welfare ranges. Benefit scale remains the dominant welfare driver, followed by contribution-to-stock conversion, reputation, demand-to-pressure conversion, and subsidy. Reputation, subsidy, and penalty can drive the classified free-riding ratio close to zero, but their pressure effects differ. This supports the multi-criteria interpretation: correcting private incentives is necessary, but pressure control also depends on the dynamic stock-pressure channel.

**Table 10:** One-dimensional sensitivity summary in the critical-infrastructure synthetic scenario.

| Parameter | Tested range | Welfare range | Free-riding range | Pressure range |
|---|---|---|---|---|
| benefit_scale | [0.810, 1.890] | 35.90–102.16 | 0.257–0.286 | 0.478–0.503 |
| alpha | [0.468, 1.092] | 48.82–76.01 | 0.282–0.286 | 0.444–0.510 |
| reputation | [0.000, 0.600] | 68.63–86.86 | 0.000–0.286 | 0.254–0.489 |
| lambda | [0.264, 0.616] | 61.76–78.00 | 0.257–0.286 | 0.301–0.613 |
| subsidy | [0.000, 0.600] | 68.63–83.78 | 0.000–0.286 | 0.205–0.489 |
| eta | [0.180, 0.420] | 62.59–74.56 | 0.257–0.286 | 0.460–0.522 |
| penalty | [0.000, 0.600] | 68.63–79.64 | 0.004–0.286 | 0.421–0.489 |

## 10.2 Subsidy-Penalty Response Surface

We also conduct a two-dimensional grid experiment for subsidy and penalty intensity. Under the updated capacity-saturating dynamics, the highest-welfare grid point is no longer pure penalty. The best tested combination is subsidy 0.150 and penalty 0.600, with average tail welfare 84.232 and zero classified free riding. The top five points all combine a positive subsidy with a relatively strong penalty. This result suggests that, in the updated model, moderate reward and strong deterrence are complementary rather than perfect substitutes.

**Table 11:** Top welfare points in the subsidy-penalty response-surface experiment.

| Subsidy | Penalty | Welfare | Free riding | Pressure | Policy cost | Effort Gini |
|---|---|---|---|---|---|---|
| 0.150 | 0.600 | 84.232 | 0.000 | 0.297 | 2.355 | 0.379 |
| 0.150 | 0.525 | 84.169 | 0.000 | 0.303 | 2.253 | 0.381 |
| 0.150 | 0.450 | 84.083 | 0.000 | 0.310 | 2.152 | 0.383 |
| 0.150 | 0.375 | 83.976 | 0.000 | 0.316 | 2.050 | 0.388 |
| 0.150 | 0.300 | 83.852 | 0.000 | 0.323 | 1.950 | 0.392 |

Figure 13 visualizes the same subsidy-penalty grid as a response surface. Welfare is the height of the surface, and color represents average maintenance pressure. The red marker denotes the highest-welfare tested grid point. The response surface should not be read as a recommendation to maximize punishment or subsidy mechanically; it is a controlled sensitivity experiment within the implemented cost and behavioral-response assumptions.

The grid also clarifies the asymmetric roles of subsidy and penalty in this scenario. Subsidy rewards contributors and can raise effort broadly, while penalty is more targeted around low-effort high-benefit behavior. The updated optimum uses both instruments, indicating that reward and deterrence can be complementary in a capacity-saturating stock-pressure system. However, the pressure color field does not perfectly follow the welfare height, which means that incentive correction does not fully replace backlog reduction or capacity-oriented support.

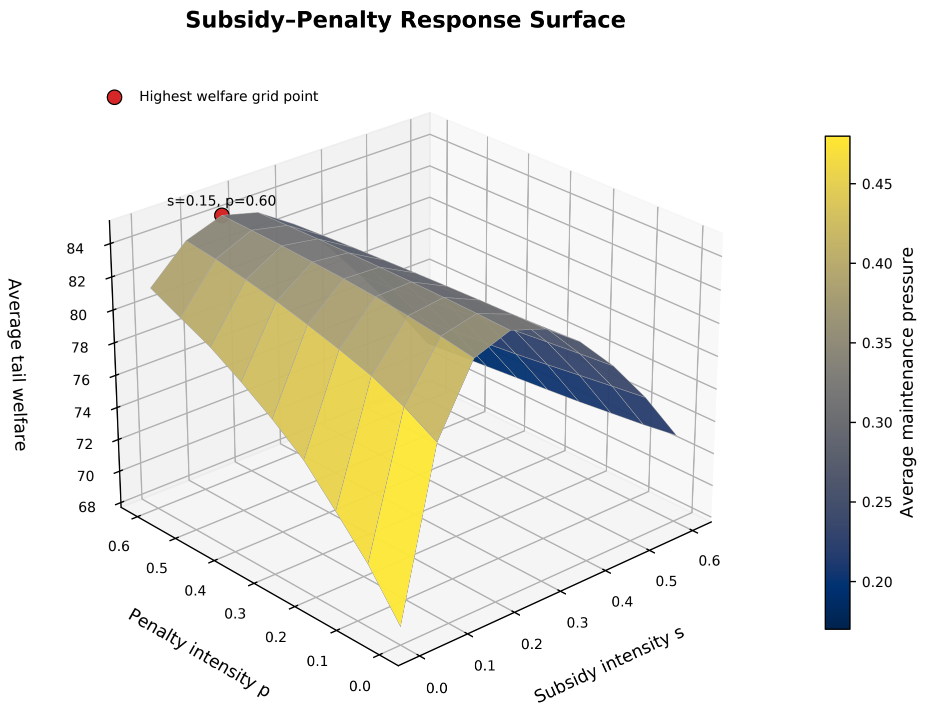


**Figure 13:** Three-dimensional subsidy-penalty response surface in the critical-infrastructure synthetic scenario. Welfare is the vertical axis, and color represents average maintenance pressure. The red marker denotes the highest-welfare tested grid point.

## 10.3 Robustness under Perturbed Scenarios

Robustness analysis perturbs the critical-infrastructure scenario by changing decay, cost, and demand conditions. Table 12 compares baseline and combined-portfolio outcomes. Under every tested perturbation, the combined portfolio improves welfare and reduces classified free riding to zero. The demand-shock case still has high pressure under intervention, but the pressure is lower than the corresponding baseline pressure. This strengthens the interpretation that a portfolio can provide both behavioral correction and partial pressure relief, although structural demand growth still requires capacity-oriented management.

**Table 12:** Robustness comparison between baseline and combined portfolio under perturbed synthetic settings.

| | Welfare | | Free-riding ratio | | Pressure | |
|---|---|---|---|---|---|---|
| Variant | Baseline | Combined | Baseline | Combined | Baseline | Combined |
| Demand shock | 57.72 | 78.14 | 0.257 | 0.000 | 0.684 | 0.532 |
| High cost | 66.34 | 84.57 | 0.286 | 0.000 | 0.505 | 0.286 |
| High decay | 67.00 | 84.25 | 0.286 | 0.000 | 0.492 | 0.261 |
| Low cost | 70.38 | 85.29 | 0.279 | 0.000 | 0.491 | 0.240 |
| Low decay | 69.56 | 85.66 | 0.286 | 0.000 | 0.507 | 0.268 |

### 10.4 Monte Carlo Uncertainty Analysis

Monte Carlo experiments perturb parameters and rerun the critical-infrastructure scenario under baseline, reputation, and combined-portfolio policies. Table 13 and Figure 14 show that both interventions shift the welfare distribution upward and sharply reduce free riding. Under the updated capacity-saturating dynamics, the combined portfolio has the highest mean welfare and almost zero classified free riding, while reputation delivers nearly the same mean welfare at a much lower average policy cost. The uncertainty analysis therefore supports a conditional recommendation: use reputation when the priority is low-cost incentive correction, and use the combined portfolio when pressure control and system protection justify higher intervention cost.

**Table 13:** Monte Carlo uncertainty summary under synthetic perturbations.

| Policy | Welfare mean | Welfare std | Final $G$ mean | Free-riding mean | Cost mean |
|---|---|---|---|---|---|
| Baseline | 76.53 | 15.82 | 0.830 | 0.253 | 0.00 |
| Combined portfolio | 93.92 | 13.68 | 1.287 | 0.003 | 4.59 |
| Reputation | 93.53 | 13.70 | 1.171 | 0.017 | 0.12 |

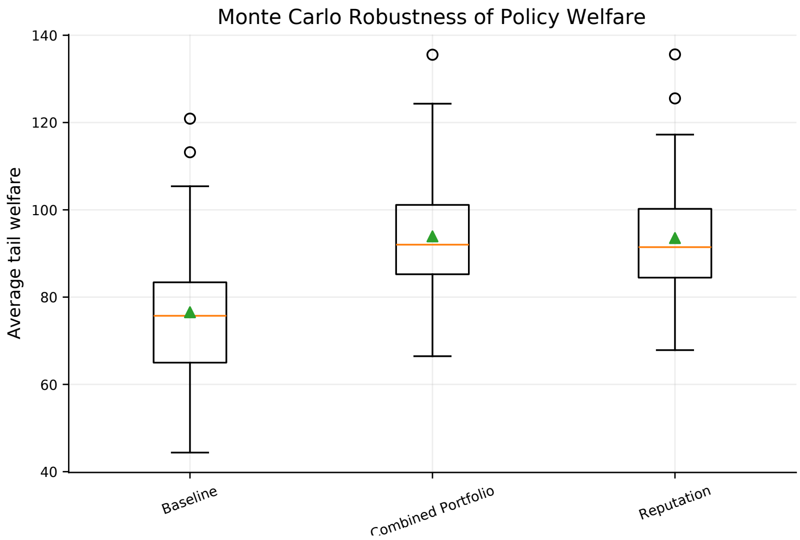


**Figure 14:** Welfare distribution under Monte Carlo synthetic perturbations. The boxplot shows that reputation and combined portfolio policies shift the welfare distribution upward relative to the baseline; the corresponding free-riding means are reported in Table 13.

Figure 14 complements the deterministic policy comparison. The deterministic table reports one controlled setting, while Monte Carlo runs show how outcomes vary after parameter perturbations. Baseline welfare has a wide spread, which means that the uncontrolled system is sensitive to parameter uncertainty. Reputation and combined portfolio policies shift the welfare distribution upward and reduce free riding, suggesting that the qualitative intervention conclusion is not driven only by one exact parameter setting.

Figure 15 adds the time dimension to the uncertainty analysis. The uncertainty bands show that the baseline remains more volatile across the simulated horizon, whereas reputation and combined portfolio policies keep the welfare path in a higher and narrower region after the initial adjustment. This is why the Monte Carlo result is interpreted as both a level improvement and a robustness improvement.

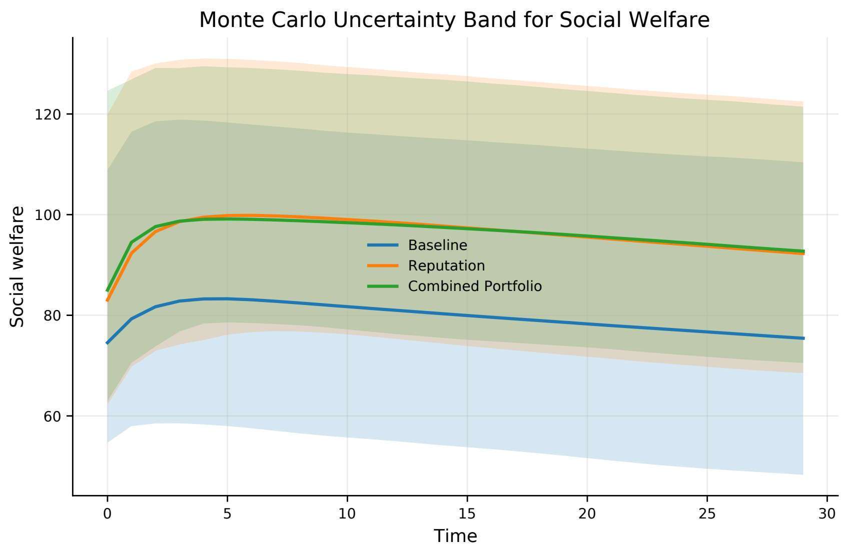


**Figure 15:** Time-varying welfare uncertainty bands under Monte Carlo synthetic perturbations. Lines show mean welfare paths and shaded bands show simulation uncertainty.

The two improved policies have different interpretations. Reputation has nearly the same mean welfare as the combined portfolio and very low cost, so it is a cost-effective behavioral instrument. The combined portfolio has the highest mean welfare and higher cost, but it almost eliminates free riding and provides stronger pressure-oriented protection. Therefore, uncertainty analysis supports a conditional recommendation: use reputation when the goal is low-cost incentive correction, and use combined portfolios when the system faces severe maintenance pressure or high public-good importance.

# 11 Policy Recommendations

The numerical experiments suggest that governance should be selected by scenario conditions rather than by a single universal instrument. The following rules translate the model outputs into operational recommendations. They should be read as mechanism-based decision rules under the synthetic DSPF framework, not as empirical prescriptions for a specific public-good system.

### 11.1 Recommendation 1: Use Low-Cost Incentive Correction Before Heavy Intervention

The individual-rational and social-planner comparison indicates that decentralized individual rationality systematically under-provides the public good under the synthetic scenario settings. Therefore, the first governance step should be to correct private marginal incentives. Reputation mechanisms, targeted penalties, or subsidies can increase effort by raising private marginal benefit or lowering effective private cost.

For low-pressure voluntary settings, the recommended decision rule is: start with reputation or light subsidy, monitor the free-riding ratio and effort inequality, and avoid large direct budget injections unless maintenance pressure begins to accumulate. This rule is consistent with the policy comparison, where reputation produces nearly the same long-run welfare as the combined portfolio at much lower cost.

### 11.2 Recommendation 2: Switch to Portfolio Governance under High Maintenance Pressure

When maintenance pressure is high, free-riding reduction alone is not enough. The critical-infrastructure simulations show that the combined portfolio creates the largest pressure reduction and the highest deterministic welfare, although its policy cost is higher. For infrastructure-like public goods, decision makers should combine incentive correction with backlog reduction and direct budget support when pressure relief is the dominant objective.

Operationally, a portfolio policy is justified when two signals appear together: the free-riding ratio remains high and the pressure trajectory continues to rise even after incentive correction. In that case, a pure reputation or penalty mechanism may improve effort but may not remove accumulated backlog. The model therefore supports a staged policy design: first correct incentives, then add pressure-relief instruments if the stock-pressure system remains fragile.

### 11.3 Recommendation 3: Avoid One-Size-Fits-All Policy

The model suggests that policy should depend on scenario conditions. In small-volunteer settings, heavy interventions may reduce net welfare because policy cost can outweigh additional benefit. In rapid-growth settings, subsidy and targeted incentive correction can help scale contribution, while matching alone should be monitored because it may leave free riding and pressure unresolved. In burnout-prone settings, simply forcing more contribution may worsen burden; a balanced portfolio with support and pressure relief is more appropriate.

The Pareto results further suggest that policy makers should compare portfolios by welfare, cost, free-riding reduction, fairness, pressure, and stability. In practice, the model can be used as a screening workflow: identify the dominant risk, choose a low-cost incentive correction if the system is still stable, and escalate to a mixed portfolio only when pressure or reliability risk dominates.

## 12 Team Member Contributions

This study was completed independently. The author was responsible for problem interpretation, mechanism design, mathematical modeling, computational modeling, numerical experiments, figure generation, result interpretation, and final manuscript writing.

**Table 14:** Scenario-based managerial decision rule suggested by the DSPF experiments.

| Scenario type | Dominant risk | Suggested governance response |
|---|---|---|
| Small volunteer | Low pressure but uneven contribution | Use light reputation or matching; avoid costly heavy intervention. |
| Rapid growth | Demand expansion and pressure accumulation | Support scalable contribution through subsidy or matching, then monitor pressure. |
| Critical infrastructure | High public value and maintenance burden | Use reputation or penalty for incentive correction; add portfolio tools if pressure dominates. |
| Burnout-prone | High cost, high pressure, and contributor fatigue | Combine incentives with backlog reduction and budget support rather than only forcing effort. |

# 13 Discussion: Strengths, Limitations, and Extensions

## 13.1 Strengths

First, the model directly corresponds to the problem requirements: it includes multiple agents, dynamic stock, natural decay, individual benefit, contribution cost, individual rationality, the social-planner benchmark, and intervention mechanisms. Second, it separates mechanism modeling from empirical claims by using explicit synthetic labels and a data audit. Third, it compares Nash-style individual rationality and a social-planner benchmark in the same state space, making the free-riding gap measurable. Fourth, policy evaluation is multi-criteria rather than single-objective. Fifth, sensitivity, robustness, Monte Carlo, and Pareto analyses reduce dependence on one parameter setting.

## 13.2 Limitations

The main limitation is that the model is synthetic-only. It is useful for mechanism exploration, but it cannot estimate the magnitude of free riding in a specific real-world system. Agent utility is stylized and does not include social norms, communication, network effects, legal constraints, or repeated-game learning. The Nash-style solution is a numerical equilibrium proxy rather than a proven unique Nash equilibrium. The social-planner benchmark is stage-wise and not an infinite-horizon global optimum. Policy acceptability and implementation feasibility are simplified into cost terms.

## 13.3 Future Work

Future work can calibrate the model with real public-goods data when reliable observations are available, include network interactions among agents, model learning and reputation accumulation over repeated periods, and add explicit institutional constraints. A hybrid empirical-simulation strategy would be natural once trustworthy data on contribution, public-good quality, and maintenance cost are available. Possible calibration proxies include open-source maintenance records, community service participation logs, infrastructure maintenance budgets, or environmental governance indicators, but such proxies should be used cautiously because they observe only part of the public-good mechanism.

## 13.4 Reproducibility and Auditability Note

The numerical study is designed to be reproducible at the level of scenario definitions, random seed, policy settings, and displayed precision. All reported scenarios are synthetic, the random seed is fixed at 20260614, and the same parameter set is used for the benchmark comparison, policy evaluation, sensitivity analysis, robustness tests, Monte Carlo experiments, and Pareto search. The tables report rounded values, so only machine-precision changes are expected under independent reruns. This auditability boundary is important: reproducibility supports internal consistency of the mechanism study, but it does not turn the synthetic scenarios into empirical measurements of a particular public-good system.

# 14 Conclusion

This paper constructs a simulation-based dynamic multi-agent model for free riding in public-good provision. The model links heterogeneous agents, contribution, public-good stock, maintenance pressure, demand feedback, individual rationality, the social-planner benchmark, and policy intervention. Under controlled synthetic scenario settings, the numerical results indicate that Nash-style individual rationality under-provides the public good relative to the stage-wise social-planner benchmark. The burnout-prone scenario has the largest free-riding gap, under-provision ratio, and welfare-loss ratio, while the critical-infrastructure scenario has the largest one-step maintenance-pressure gap.

Policy experiments suggest that well-matched interventions can reduce free riding and improve welfare, but their effects depend on scenario conditions and differ in cost, pressure relief, fairness, and stability. Reputation is cost-effective in the critical-infrastructure scenario, while the combined portfolio is more suitable when maintenance pressure and system protection are central concerns. Sensitivity, robustness, Monte Carlo, and Pareto analyses reinforce the main qualitative conclusion: public-good governance should not rely on a single instrument, and policy recommendations should be scenario-specific.

The practical value of the DSPF framework is diagnostic. It provides a reproducible workflow for asking whether a public-good system is primarily limited by weak private incentives, accumulated maintenance pressure, excessive policy cost, or unequal contribution burden. All conclusions remain conditional on the synthetic simulation design. The model does not claim empirical calibration; instead, it offers a mechanism-oriented framework for understanding how free riding can emerge dynamically and how policy instruments can be compared systematically.

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